PAKISTAN ORDNANCE FACTORIES (POF) WAH CANTT

Internship Design Proposal

Internship Period: 03-08-2026 to 14-09-2026

Name	Atif Ali Shah
Reg No	FA23-BAI-004
Program	BS Artificial Intelligence
University	COMSATS University Islamabad, Wah Campus
Organization	Pakistan Ordnance Factories (POF) Wah Cantt, IT Department (CO-4)

Career Goal

My career goal is to become an AI and data professional with a strong foundation in databases. Understanding how data is stored, secured and managed is essential for building reliable AI systems, so this internship in Database Administration supports my intended path in data science and AI/ML engineering.

Learning Objectives

To understand the role of a DBA, set up a virtualized Oracle Linux environment, install and configure Oracle Database 12c, work with SQL Developer and Toad, design and implement a normalized database, and learn backup/recovery (RMAN), database links, security, monitoring and troubleshooting.

Organization Overview & Role

Pakistan Ordnance Factories (POF) Wah Cantt is a major defense manufacturing organization. I am attached to the IT Department (CO-4), Database Administration section, as an intern DBA. My role is to assist in database setup, administration, documentation and troubleshooting under the Site Supervisor.

Key Activities

Coding • Testing • Documentation • Technical Support • Research

Tools / Technologies

VMware Workstation, Oracle Linux, Oracle Database 12c, Oracle Universal Installer, SQL Developer, Toad for Oracle, SQL/PL-SQL, RMAN.

Expected Deliverables

Working Oracle 12c installation on a virtual machine, a boutique management database (tables, constraints, sequences, queries), database link and backup/recovery practicals, six weekly reports and a final internship report.

Academic Preparation

Database Systems, SQL, Data Structures, Operating Systems, Software Engineering and AI coursework, which gave me the base for understanding database concepts and Linux-based environments.